

MACHKHAN (Candesartan Cilexetil 8mg/ Amlodipine Besylate 5mg) Film-Coated Tablets

MACHKHAN (Candesartan Cilexetil 16mg/ Amlodipine Besylate 5mg) Film-Coated Tablets

1. Brand or Product Name

MACHKHAN (Candesartan Cilexetil 8mg/ Amlodipine Besylate 5mg) Film-Coated Tablets

MACHKHAN (Candesartan Cilexetil 16mg/ Amlodipine Besylate 5mg) Film-Coated Tablets

2. Name and Strength of Active Substance(s)

MACHKHAN (Candesartan Cilexetil 8mg/ Amlodipine Besylate 5mg) Film-Coated Tablets (185.0 mg) contains Candesartan cilexetil(EP) 8.00 mg / Amlodipine besylate(USP) 6.935 mg(5 mg as Amlodipine)

MACHKHAN (Candesartan Cilexetil 16mg/ Amlodipine Besylate 5mg) Film-Coated Tablets (277.5 mg) contains Candesartan cilexetil(EP): 16.00 mg / Amlodipine besylate(USP) 6.935 mg(5 mg as Amlodipine)

3. Product Description

MACHKHAN (Candesartan Cilexetil 8mg/ Amlodipine Besylate 5mg) Film-Coated Tablets: Pink-colored round film-coated tablet with convex on both sides

MACHKHAN (Candesartan Cilexetil 16mg/ Amlodipine Besylate 5mg) Film-Coated Tablets: Yellow-colored round film-coated tablet with convex on both sides

4. Pharmacodynamics and Pharmacokinetics

4.1. Pharmacodynamics properties

Pharmacotherapeutic group: Angiotensin II antagonists and calcium channel blockers, ATC code C09DB07.

Mechanism of action

This drug is a combination of an Angiotensin II antagonists, candesartan cilexetil, and calcium channel blockers, amlodipine besylate. The combination of these active ingredients has an additive antihypertensive effect, reducing blood pressure to a greater degree than either component alone

- Amlodipine

Amlodipine is a calcium ion influx inhibitor (slow channel blocker on calcium ion antagonist) and inhibits the transmembrane influx of calcium ions into cardiac and vascular smooth muscle.

The mechanism of the antihypertensive action of amlodipine is due to a direct relaxant effect on vascular smooth muscle. The precise mechanism by which amlodipine relieves angina has not been fully determined, but amlodipine reduces total ischemic burden by the following two actions:

1) Amlodipine dilates peripheral arterioles and thus reduces the total peripheral resistance (afterload) against which the heart works. Since the heart rate remains stable, this unloading of the heart reduces myocardial energy consumption and oxygen requirements.

2) The mechanism of action of amlodipine also probably involves dilatation of the main coronary arteries and coronary arterioles, both in normal and ischemic regions. This dilation increases myocardial oxygen delivery in patients with coronary artery spasm (Prinzmetal's or variant angina) and blunts smoking-induced coronary vasoconstriction.

- Candesartan cilexetil

Angiotensin II is the primary vasoactive hormone of the renin-angiotensin-aldosterone system and plays a role in the pathophysiology of hypertension, heart failure and other cardiovascular disorders. It also has a role in the pathogenesis of end organ hypertrophy and damage. The major physiological effects of angiotensin II, such as vasoconstriction, aldosterone stimulation, regulation of salt and water homeostasis and stimulation of cell growth, are mediated via the type 1 (AT₁) receptor.

Candesartan cilexetil is a prodrug suitable for oral use. It is rapidly converted to the active drug, candesartan, by ester hydrolysis during absorption from the gastrointestinal tract. Candesartan is an angiotensin II receptor

antagonist, selective for AT₁ receptors, with tight binding to and slow dissociation from the receptor. It has no agonist activity.

Candesartan does not inhibit ACE, which converts angiotensin I to angiotensin II and degrades bradykinin. There is no effect on ACE and no potentiation of bradykinin or Substance P. In controlled clinical trials comparing candesartan with ACE inhibitors, the incidence of cough was lower in patients receiving candesartan cilexetil. Candesartan does not bind to or block other hormone receptors or ion channels known to be important in cardiovascular regulation. The antagonism of the angiotensin II (AT₁) receptors results in dose related increases in plasma renin levels, angiotensin I and angiotensin II levels, and a decrease in plasma aldosterone concentration.

Results of clinical studies in patients who were non-responders to monotherapy.

MACHKHAN showed superior antihypertensive efficacy as both treatment response rate and treatment goal achievement rate were higher in patients with essential hypertension not adequately controlled by candesartan cilexetil monotherapy or amlodipine monotherapy.

The results suggest that in essential hypertension patients not controlled by monotherapy, the combination of candesartan cilexetil/amlodipine can lower blood pressure effectively, providing a potential opportunity for achieving better blood pressure control.

4.2. Pharmacokinetics properties

- Amlodipine

Absorption

After oral administration of therapeutic doses, amlodipine is well absorbed with peak blood levels between 6 and 12 hours post-dose. Absolute bioavailability has been estimated to be between 64% and 80%. The volume of distribution is approximately 21 L/kg. Absorption of amlodipine is unaffected by consumption of food.

In vitro studies have shown that approximately 97.5% of circulating amlodipine is bound to plasma proteins.

Biotransformation/Elimination

The terminal plasma elimination half-life is about 35 to 50 hours and is consistent with once-daily dosing. Steady-state plasma levels are reached after 7 to 8 days of consecutive dosing. Amlodipine is extensively metabolized by the liver to inactive metabolites, with 10% of the parent compound and 60% of metabolites excreted in the urine.

Use in the Elderly

The time to reach peak plasma concentrations of amlodipine is similar in elderly and younger subjects. Amlodipine clearance tends to be decreased with resulting increases in AUC and elimination half-life in elderly patients. Increases in AUC and elimination half-life in patients with CHF were as expected for the patient age group studied.

Use in Pediatrics

In one clinical chronic exposure study, 73 hypertensive pediatric patients aged 12 months to less than or equal to 17 years received amlodipine at an average daily dose of 0.17 mg/kg. Clearance for subjects with median weight 45 kg was 23.7 L/h and 17.6 L/h for males and females, respectively. This is a similar range to the published estimates of 24.8 L/h in a 70 kg adult. The average estimate for volume of distribution for a 45 kg patient was 1130 L (25.11 L/kg). Maintenance of the blood pressure effect over the 24-hour dosing interval was observed with little difference in peak and trough variation effect. When compared to historical adult pharmacokinetics, the parameters observed in this study indicate that once-daily dosing is appropriate.

- Candesartan cilexetil

Absorption and Distribution

Following oral administration, candesartan cilexetil is converted to the active substance candesartan. The absolute bioavailability of candesartan is approximately 40% after an oral solution of candesartan cilexetil. The relative bioavailability of the tablet formulation compared with the same oral solution is approximately 34% with very little variability. The estimated absolute bioavailability of the tablet is therefore 14%. The mean peak serum concentration (C_{max}) is reached 3-4 hours following tablet intake. The candesartan serum concentrations increase linearly with increasing doses in the therapeutic dose range. No gender related differences in the pharmacokinetics of candesartan have been observed. The area under the serum concentration *versus* time curve (AUC) of candesartan is not significantly affected by food.

Candesartan is highly bound to plasma protein (more than 99%). The apparent volume of distribution of candesartan is 0.1 l/kg.

Metabolism and elimination

Candesartan is mainly eliminated unchanged via urine and bile and only to a minor extent eliminated by hepatic metabolism. The terminal half-life of candesartan is approximately 9 hours. There is no accumulation following multiple doses.

Total plasma clearance of candesartan is about 0.37 ml/min/kg, with a renal clearance of about 0.19 ml/min/kg. The renal elimination of candesartan is both by glomerular filtration and active tubular secretion. Following an oral dose of ^{14}C -labelled candesartan cilexetil, approximately 26% of the dose is excreted in the urine as candesartan and 7% as an inactive metabolite while approximately 56% of the dose is recovered in the faeces as candesartan and 10% as the inactive metabolite.

Pharmacokinetics in special populations

In the elderly (over 65 years) both C_{max} and AUC of candesartan are increased by approximately 50% and 80%, respectively in comparison to young subjects. However, the blood pressure response and the incidence of adverse events are similar after a given dose of candesartan in young and elderly patients.

In patients with mild to moderate renal impairment C_{max} and AUC of candesartan increased during repeated dosing by approximately 50% and 70%, respectively, but $t_{1/2}$ was not altered, compared to patients with normal renal function. The corresponding changes in patients with severe renal impairment were approximately 50% and 110%, respectively. The terminal $t_{1/2}$ of candesartan was approximately doubled in patients with severe renal impairment. The AUC of candesartan in patients undergoing hemodialysis was similar to that in patients with severe renal impairment.

In patients with mild to moderate hepatic impairment, there was a 23% increase in the AUC of candesartan.

5. Indication

Essential hypertension not adequately controlled with Candesartan Cilexetil or Amlodipine monotherapy.

6. Recommended Dosage

The recommended dose of this drug is to be taken once daily with or without food. It is recommended to take this drug with water. It recommends to take this drug same time (e.g. Morning) if possible.

Individual dose titration with the components (i.e. Candesartan Cilexetil or amlodipine) is recommended before taking this drug, but if blood pressure is not controlled by monotherapy for the individual components as follows, direct switch to this drug may be considered.

- 8/5 mg: Patient whose blood pressure is not adequately controlled on candesartan cilexetil 8 mg or amlodipine 5 mg monotherapy
- 16/5 mg: Patient whose blood pressure is not adequately controlled on candesartan cilexetil 16 mg or amlodipine 5 mg monotherapy

For convenience, patients receiving candesartan cilexetil and amlodipine from separate tablets/capsules may be

switched to this drug (containing the same component doses).

- **Patients with Renal Impairment**
The starting dose is 4 mg as candesartan cilexetil per day in patient with renal impairment. It is not recommended to administer this drug during initial treatment.
- **Patients with Hepatic Impairment**
The starting dose of 2 mg as candesartan cilexetil once daily is recommended in patients with mild to moderate hepatic impairment. It is not recommended to administer this drug during initial treatment.
- **Pediatric Population**
It is not recommended to administer this drug, because the safety and efficacy is not established for children and adolescents ≤ 18 years.
- **Elderly**

In general, administration this drug to elderly patient should be cautious since the excessive blood pressure decrease in elderly has a risk of cerebral infarction.

7. Route of Administration

For oral administration only

8. Contraindications

- 8.1 Patients having hypersensitivity with this drug or dihydropyridine type drug.
- 8.2 Pregnancy or women who may be pregnant and Nursing mother.
- 8.3 Patients with severe renal impairment and/or cholestasis.
- 8.4 The combination with aliskiren-containing medicines in patients with diabetes mellitus or with moderate to severe renal impairment ($GFR < 60 \text{ ml/min/1.73 m}^2$).
- 8.5 Patients with hereditary angioedema, history of angioedema when treatment with ACE Inhibitor or Angiotensin II receptor antagonist.
- 8.6 Patients with primary hyperaldosteronism (their Renin-angiotensin-aldosterone disorders(RAAS) is not activated. Don't administrate this drug)
- 8.7 Patients with severe aortic stenosis
- 8.8 Patients with shock

9. Warnings and Precautions

9.1 Warning

When used in pregnancy, drugs that act directly on the renin-angiotensin-aldosterone disorders(RAAS) can cause injury and even death to the fetus and newborn. And this cases also were reported taking ACE inhibitor worldwide. When pregnancy is detected, this drug should be discontinued as soon as possible. (See 11. Pregnancy and Lactation)

9.2 Precautions

The drug is administered with caution to the following patients

- Elderly patients.
- Patients with hyperkalemia or easy to elevate potassium level in serum.
- Patients with mild to moderate hepatic impairment (ex. Patient with cirrhosis)
- Patients with severe hypotension.
- Patients with aortic and mitral valve stenosis or obstructive hypertrophic cardiomyopathy
- Patients with ischemic cardiac disease, ischemic cardiovascular disease, and cerebrovascular

disorder (excessive blood pressure decrease may occur myocardial infarction or stroke due to cerebral blood flow failure)

- Patients with unilateral or bilateral renal artery stenosis (concern about rapidly reducing renal function due to a decrease in renal blood flow or a decrease in glomerular filtration pressure)
- Patients with Intravascular volume depletion.
- Patients with renal impairment (concern about reducing renal function due to excessive pressure)
- Patients who have undergone renal transplant recently. (There is no evidence to use this drug)
- Patients with renal failure requiring dialysis.
- Dual blockade of the renin-angiotensin-aldosterone disorders(RAAS)
: The concomitant use of other medicinal products, such as angiotensin II receptor blockers(ARB)s, ACE-inhibitors or aliskiren, what impact on the renin-angiotensin-aldosterone disorders(RAAS) is not recommended.
- Children and adolescents ≤ 18 years

9.3 General Precaution

- Patients with renal artery stenosis

Should be used with caution for patient with unilateral or bilateral renal artery stenosis since as like other Renin-angiotensin-aldosterone system agents (such as ACE inhibitor) blood urea and serum creatinine may increase in such patients. Similar effect may occur by angiotensin II receptor antagonist.

- Hypotension

Since symptomatic hypotension may occur in patient with heart failure and intravascular volume decrease (e.g. Patients receiving high doses of diuretics) as with other drugs acting on renin-angiotensin-aldosterone system, correction of this condition prior to administration of this drug. In addition, patients who are receiving angiotensin II receptor antagonist, hypotension may occur during anesthesia and surgery due to renin-angiotensin-aldosterone system. Very rarely, severe hypotension may be induced, requiring intravenous administration of fluids or vasoconstrictors. It is recommended not to administer this drug for 24 hours before surgery.

- Hyperkalemia

Hyperkalemia may occur in patients with renal failure who are receiving this drug, patients with renal impairment and diabetes. It is recommended to regular monitoring during treatment. Especially, the combination of an ACE inhibitor, a potassium-sparing diuretic (e.g. spironolactone) need careful monitoring. And be caution since the symptoms may be severe in patient with hyperkalemia.

- Heart Failure

The risk of adverse reactions especially renal impairment and hyperkalemia may increase in patients with heart failure who are receiving this drug with ACE inhibitor. And administration of this drug in the patients with heart failure may induce renal function decrease and anemia hence when administering to the following patients while thoroughly observing patient's condition such as blood pressure, renal function, anemia index (hemoglobin etc.) and then start treatment and increase dose carefully.

① Patients on hemodialysis

② Patients on strict salt restriction

③ Patients taking diuretics (in particular, who have recently started taking diuretic recently)

- ④ Patients with hyponatremia
- ⑤ Renal impairment
- ⑥ Patients with hypotension
- ⑦ Patients with severe chronic heart failure such as NYHA Class III

In the case of candesartan cilexetil, periodic evaluation of renal function should be performed for the elderly aged 75 years or older among heart failure patients or those with impaired renal function, and it is recommended to monitor serum potassium and creatinine levels when increasing the dose. In a long-term, placebo-controlled study (PRAISE-2) of amlodipine in patients with NYHA (New York Heart Association Classification) III and IV heart failure of non-ischemic aetiology, amlodipine was associated with increased reports of pulmonary oedema despite no significant difference in the incidence of worsening heart failure as compared to placebo.

- Renal impairment and kidney transplantation

In the case of candesartan cilexetil, the renal function may be changed in sensitive patients. Regular monitoring of potassium levels and creatinine is advised in patients with renal impairment. Clinical experience in patients with severe or end-stage renal impairment (creatinine clearance <15 mL/min) is limited. There is no experience of using this drug in patients who have recently undergone a kidney transplant.

- Hepatic impairment

As with all calcium antagonist, the half-life of amlodipine is longer in patients with hepatic impairment, and dosage recommendations have not been established for these patients. Therefore, caution should be exercised when administering this drug to these patients.

- Hemodialysis

The blood pressure may become sensitive to AT1-receptor antagonist due to decreased plasma volume and renin-angiotensin-aldosterone system activation while hemodialysis, hence when this drug is administered to these patients, the dose should be titrated with careful monitoring of blood pressure.

- Thiazide-diuretic

The concomitant use with thiazide diuretics such as hydrochlorothiazide increase action on blood pressure decrease, and this drug can be administered with other heart failure treatment, including beta blocker, non-potassium-sparing diuretics, digitalis or combination of these medicinal products.

- Patients with Hypertension

Administration of this drug may rarely cause excessive blood pressure decrease, shock, syncope, transient loss of consciousness, or deterioration of renal function. In particular, hence when this drug is administered to the following patients, start with low dosage and increase the dosage while monitoring blood pressure, renal function and condition of patient carefully.

- ① Patients on hemodialysis
- ② Patients on strict salt restriction
- ③ Patients who receiving diuretics
- ④ Patients with hypokalemia
- ⑤ Patients with Renal impairment

⑥ Patients with Heart Failure

- Patients with Ischemic disease

As with any antihypertensive agent, excessive blood pressure decrease in patients with ischemic cardiopathy or ischemic cerebrovascular disease could result in a myocardial infarction or stroke. Symptomatic hypotonia may occur in hypertension patients with intravascular volume depletion such as receiving high dose diuretics.

- In patients whose vascular tone and kidneys are primarily dependent on renin-aldosterone-angiotensin system activity (such as patients with congestive heart failure or renal disease including renal arterial infarction), treatment with other drugs that affect this system has been associated with acute hypotension, hyperazotemia, oliguria, or, rarely, acute renal failure. The possibility that a similar effect would occur with angiotensin II receptor antagonist cannot be excluded.
- In case of amlodipine, the half-life in plasma concentration is long thus after discontinue the blood pressure decrease slowly. Hence when discontinue it and start other antihypertensive agents, caution with the dosage and dosing interval while monitoring patients' condition.
- In case of amlodipine, it become on-set slowly thus it cannot be expected to be effective in unstable angina which requiring emergence treatment.

10. Interactions with Other Medicaments

10.1 There was not any significant pharmacokinetic interaction when co-administration of candesartan cilexetil and amlodipine. No drug-drug interaction studies have been performed with this drug and other medicinal products.

10.2. Drug-drug interaction of Candesartan cilexetil

- Dual blockade of the RAAS through the combined use of ACE inhibitors, ARBs or aliskiren is associated with increasing risk such as hypotension, syncope, hyperkalemia and decreased renal function (including acute renal failure). When co-administration of this drug with other medicinal products impacting on RAAS, blood pressure, renal function and electrolytic should be monitored closely. Do not use of this drug with aliskiren-containing products in patients with diabetes mellitus or renal impairment (GFR <60 ml/min/1.73 m²).
- The substances what have been evaluated in clinical pharmacokinetic studies are hydrochlorothiazide, warfarin, digoxin, oral contraceptive pill(it means ethanylestradiol /levonorgestrel), glibenclamide, nifedipine and no interactions of clinical significance have been found.
- Candesartan cilexetil is primarily eliminated in bile and urine mainly as unchanged drug. Only small amounts are eliminated by hepatic metabolism(CYP2C9). There were no effects on CYP2C9 and CYP3A3 and the effects on other CYP-450 isoenzymes has not be reported.
- The effect of blood pressure decrease can be induced by other antihypertensive agents.
- Reversible increases in serum lithium concentrations and toxicity have been reported during concomitant administration of lithium with ACE inhibitors. Similar effect may occur with angiotensin II receptor antagonists. Therefore, careful monitoring of serum lithium levels is recommended during concomitant use.
- When angiotensin II antagonists are administered simultaneously with NSAIDs(e.g. selective COX-2 inhibitors, acetylsalicylic acid (>3 g/day), and non-selective NSAIDs) attenuation of the antihypertensive effect may occur. As with ACE inhibitor, concomitant use of angiotensin II antagonists and NSAIDs may lead to an increased risk of worsening of renal function and an increase in serum potassium
- Considering experience of other agents that affect the renin-angiotensin-aldosterone system, when co-administered with potassium-sparing diuretics, (spironolactone, triamterene etc.), potassium

supplements, salt supplements containing potassium or other medicinal products that may increase potassium levels (e.g. heparin, etc.), should be careful as serum potassium concentration may increase.

- This drug bioavailability is unaffected by food ingestion.
- The efficacy and safety of taking candesartan cilexetil and co-administration of candesartan cilexetil and ACE inhibitors in patients with insufficient ACE inhibitor effect have not been established. Therefore, concomitant use should be undertaken with caution
- Since the interaction of this drug with acenocoumarol or phenprocoumon has not been studied, caution should be exercised when co-administering the above anticoagulants with this drug.
- Concomitant use of this drug with ACE inhibitors or potassium-sparing diuretics is not recommended.

10.3. Drug-drug interaction of Amlodipine

- It was safe when used amlodipine in combination with thiazide diuretics, alpha-blocker, ACE inhibitor, long-acting nitrate agents, nitroglycerin sublingual tablets, non-steroidal anti-inflammatory drugs(NSAIDs), antibiotics, oral hypoglycemic agents.
- Amlodipine do not impact on the protein binding of digoxin phenytoin, warfarin and indomethacin according to experimental data using human plasma
- Grapefruit juice: Administration of amlodipine with grapefruit or grapefruit juice is not recommended as bioavailability may be increased in some patients, resulting in increased blood pressure lowering effects.
- Cimetidine, aluminum/magnesium(antacid), sildenafil: there was no significant impact on pharmacokinetics of amlodipine.
- CYP3A4 inhibitors: In elderly, concomitant use of amlodipine 5 mg with diltiazem 180 mg raised amlodipine systemic exposure by 57%. Concomitant use of amlodipine with strong or moderate CYP3A4 inhibitors (e.g. Ketoconazole, itraconazole, ritonavir) may give rise to significant increase in amlodipine exposure more than diltiazem. Therefore, amlodipine should be administered with caution when co-administered with CYP3A4 inhibitors
- CYP3A4 inducer: Upon co-administration of known CYP3A4 inducers (e.g. Rifampicin, St. John'S Wort), the plasma concentration of amlodipine may vary. Therefore, blood pressure should be monitored and dose regulation considered both during and after concomitant medication particularly with strong CYP3A4 inducers Atorvastatin, digoxin, ethanol(alcohol), warfarin, cyclosporine:
- Atorvastatin, digoxin, ethanol(alcohol), warfarin, cyclosporine: There was no significant pharmacokinetics and pharmacodynamics effect by amlodipine
- Tacrolimus: Since there is a risk of an increase in the blood concentration of tacrolimus when co-administered with this drug, the blood concentration of tacrolimus should be monitored when taking this drug in patients treated with tacrolimus, and the dose of tacrolimus should be appropriately adjusted to avoid tacrolimus toxicity.
- Simvastatin: Multi-dose administration of amlodipine 10 mg and simvastatin 80 mg increased the exposure of simvastatin by 77% compared to simvastatin alone. In patients receiving amlodipine, the maximum daily dose of simvastatin is 20 mg

11. Pregnancy and Lactation

The safety of this drug in human pregnancy and nursing mother has not been established.

11.1 Pregnancy

Considering the mechanism of action of angiotensin II receptor antagonists, the risk to the fetus cannot be ignored. It was associated with fetal and neonatal injuries, including hypotension, hyperkalemia, neonatal cranial hypoplasia, urinalysis and/or anuria, reversible or irreversible chronic renal failure, intrauterine growth retardation, and death, in pregnant women exposed during the 2nd and 3rd trimester to certain classes of drugs acting on the renin-angiotensin-aldosterone system. Oligohydramnios, presumed to be due to decreased fetal renal function, has been reported and was associated with fetal limb contracture, craniofacial malformation, and pulmonary hypoplasia. Retrospective data showed that the use of ACE inhibitors in the first trimester was associated with a potential risk of birth defects. In addition, although it is not clear whether this is caused by this drug, there are reports of immaturity, intrauterine growth retardation, and ductus arteriosus. Like other drugs that act directly on the renin-angiotensin-aldosterone system, this drug should not be used during pregnancy or in women who are planning to become pregnant. If pregnancy is confirmed while taking the drug, stop the drug immediately. If a drug that acts on the renin-angiotensin-aldosterone system is prescribed to women of childbearing potential, the physician should be informed of the potential risks of this drug during pregnancy. If a pregnant woman has been exposed to this drug since the second trimester, fetal renal function and ultrasonography of the skull are recommended. In addition, adequate urination, hyperkalemia, and blood pressure should be closely monitored in neonates who have been exposed to this drug in utero. For amlodipine safety in pregnancy has not been. Except for delayed and prolonged labor in rats administered with amlodipine at a dose equivalent to 50 times the maximum recommended human dose, no reproductive toxicity was demonstrated in animals. Therefore, administration to pregnant women is recommended only when there are no other safe alternative drugs and when the disease itself poses a great risk to the mother and the fetus. There was no effect on fertility in rats treated with this drug.

11.2 Nursing mother:

It is not known whether candesartan cilexetil and/or amlodipine are excreted into human milk, but the secretion of candesartan cilexetil into milk was confirmed in an experiment on rat. Therefore, it is not recommended for use by nursing mother. In case of unavoidable use, lactation should be discontinued.

12. Side Effects

12.1 Candesartan cilexetil/amlodipine

The safety of this drug was evaluated by combining the clinical trials (1 of phase 2 and 2 of phase 3) in patients with essential hypertension (total 755 patients, of which 388 patients is in the group concomitant of candesartan cilexetil/amlodipine). Adverse events have been ranked under headings of frequency using the following convention: very common ($\geq 1/10$); common ($\geq 1/100$ to $< 1/10$); uncommon ($\geq 1/1,000$ to $< 1/100$); rare ($\geq 1/10,000$ to $< 1/1,000$); very rare ($< 1/10,000$); not known

Table 1. Adverse Event

Systemic Class	Organ	Frequency		
		Common	Uncommon	Very rare
Gastrointestinal disorders			hemorrhoids, Gastritis, Abdominal pain upper	Large intestine polyp, Pancreatitis, Chronic gastritis, Gastroesophageal reflux disease, Irritable bowel syndrome, Vomiting*, Constipation*, Nausea, Abdominal discomfort *
Nervous	system	Dizziness *	Head discomfort	Anosmia, Sensory

Systemic Class	Organ	Frequency		
		Common	Uncommon	Very rare
disorders		Headache *		disturbance
Investigations			Liver function test abnormal, ALT increase*, AST increase*	Blood triglycerides increased
Cardiovascular disorders			Palpitations*	
General disorders and administration site conditions			Chest discomfort *, Fatigue	Oedema peripheral*, Face oedema*, Chest pain
Musculoskeletal and connective tissue disorders			Myalgia, Arthralgia	Bursitis, Pain in extremity, Back pain
Skin and subcutaneous tissue disorders				Pruritus, Rash
Blood and lymphatic disorders				Eosinophilia *
Ear and labyrinth disorders				Hearing impaired
Eye disorders				Dry eye
Infections and infestations				Cystitis, Acute sinusitis, Pharyngitis
Injury, poisoning and procedural complications				Eye contusion, Ligament sprain, Laceration
Metabolism and nutrition disorders				Dyslipidaemia, Hypertriglyceridemia
Respiratory thoracic and mediastinal disorders				Dyspnea, Coughing
Hepatobiliary disorders				Cholecystitis
Vascular disorders				Hot flush

* The adverse reactions among the adverse events reported by subjects participating in clinical trials, determined by the researcher to be possible, probable/likely, or certainly related to the drug.

12.2 Results of Post Market Surveillance(PMS) as per Re-examination.

- As a result of PMS conducted on 799 patients for 6 years for re-examination in Korea, the incidence rate of adverse events was reported to be 8.76% (70/799 patients, 93 cases) regardless of causal relationship. There were no reports of serious adverse drug reactions from which a causal relationship could not be ruled out. In addition, unexpected adverse drug reactions for which a causal relationship cannot be ruled out are listed in the table below according to the frequency of occurrence

Frequency	Systemic Organ class	unexpected adverse drug reactions 0.25%(2/799 patients, 2 cases)
Uncommon (Less than 0.1~1%)	Clinical Examination	Blood pressure increased

- 2) As a result of PMS conducted on 714 patients for 6 years for re-examination in Korea, the incidence rate of adverse events was reported to be 2.52% (18/714 patients, 19 cases) regardless of causal relationship. There were no reports of serious adverse drug reactions from which a causal relationship could not be ruled out. In addition, unexpected adverse drug reactions for which a causal relationship cannot be ruled out are listed in the table below according to the frequency of occurrence.

Frequency	Systemic Organ class	unexpected adverse drug reactions 0.25%(2/799 patients, 2 cases)
Uncommon (Less than 0.1~1%)	Gastrointestinal disorders	Gastrointestinal pain
	Vascular disorders	orthostatic hypotension

12.3 Adverse reactions of each active pharmaceutical ingredient(API).

It is possible to show reported adverse reactions of each active pharmaceutical ingredient Even though the adverse reactions were not observed in clinical trials with this drug.

The information sated as follows is based on data from clinical trials and post marketing experience with each component of amlodipine and candesartan cilexetil.

- Amlodipine

- 1) Amlodipine is well tolerance. The very common adverse reactions from placebo clinical trial in patients with hypertension and angina are follows.
 - (1) Vascular disorders: Hot flush
 - (2) General disorders: Fatigue, Edema,
 - (3) Cardiac disorders: Palpitation
 - (4) Central and peripheral nervous disorders: Dizziness, Headache, Drowsiness
 - (5) Gastrointestinal disorders: Abdominal pain, Nausea
- 2) The adverse reactions observed rarely post marketing are follows
 - (1) General disorders: Asthenia, Malaise, Pain, Weight increase/ Weight decrease
 - (2) Vascular disorders: Hypotension, vasculitis
 - (3) Nervous disorders: Hypertonia, Hypoesthesia/Paranesthesia, Peripheral neuropathy, Syncope, Dysgeusia, Tremor, Extraparalymidal syndrome
 - (4) Reproductive disorders: Impotence, Gynecomastia
 - (5) Gastrointestinal disorders: Change of bowel habit, Dry mouth, Dyspepsia (Gastritis), Gingival hyperplasia, Pancreatitis, Vomiting.
 - (6) Metabolism and nutrition disorders: Hyperglycemia
 - (7) Musculoskeletal disorders: Arthralgia, Back pain, Muscle spasm, Myalgia
 - (8) Blood and lymphatic disorders: Leukopenia, Thrombocytopenia.
 - (9) Psychiatric disorders: Insomnia, Mood swings
 - (10) Respiratory disorders: Cough, Dyspnea, Rhinitis.
 - (11) Skin and subcutaneous tissue disorders: Alopecia, Hyperhidrosis, Purpura, Skin discoloration, Urticaria, Toxic Epidermal Necrolysis

- (12) Sensory disorders: Tinnitus, Dysopsia
 - (13) Urinary disorders: Urinary frequency increase/Urinary frequency decrease, Nocturia
 - (14) Hepatobiliary disorders: Hepatitis, jaundice, Hepatic enzymes increased was rarely reported and it is mostly consistent with cholestasis. Although some cases severe requiring hospitalization have been reported to be related to the use of amlodipine, in most cases the causal relationship to amlodipine is unclear. Very limited clinical data are available regarding amlodipine administration in patients with hepatic impairment
 - (15) Allergic reactions including pruritus, rash, angioedema and Erythema multiforme were reported rarely.
- 3) The following adverse reactions were reported rarely like other Calcium channel blocker(CCB), and it was not possible to distinguish whether these were caused by an underlying disease or a drug: Myocardial infarction, Arrhythmias (including bradycardia, ventricular tachycardia, atrial fibrillation), chest pain.
- 4) The following adverse reactions were observed.
- (1) Cardiovascular disorders: Occasionally blood pressure decrease, sino-atrial block or atrio-ventricular block and rarely abdominal discomfort may occur.
 - (2) Gastrointestinal disorders: Occasionally epigastric pain, diarrhea, loose stool and constipation may occur.
 - (3) Skin disorders: Rarely erythromelalgia, rash maculopapular may occur
 - (4) Others: Occasionally heaviness of head, feeling hot, decreased glucose tolerance, weakness.
- Candesartan cilexetil
 - 1) Essential hypertension

In general, adverse reactions were mild and temporary, and the incidence rate independent of dose, age and gender. The cases of discontinuing the treatment was similar between the group taking this medicine and placebo.

(1) The following adverse reactions occurred at least $\geq 1\%$ in candesartan cilexetil group and the higher rate than placebo group as per results of clinical trials.

 - ① General disorders: dizziness
 - ② Musculoskeletal disorders: Back pain
 - ③ Respiratory disorder: Respiratory infection (Upper respiratory tract infection, pharyngitis, Rhinitis)

(2) The following adverse reactions occurred $\geq 1\%$ but same or higher rate in a group taking this drug than placebo as per the results of clinical trials: Fatigue, peripheral edema, Chest pain, Headache, Bronchitis, Cough, Sinusitis, Nausea, Abdominal pain, Diarrhea, Vomiting, Arthralgia, Proteinuria

(3) According to clinical trials, the following adverse reactions reported as $\geq 0.5\%$ occurrence rate regardless of the causal relationship with this drug.

 - ① General disorder: Asthenia, Fever
 - ② Central and peripheral nervous disorders: Paresthesia, Dizziness

- ③ Gastrointestinal disorders: Indigestion, Gastroenteritis
- ④ Cardiovascular disorders: Tachycardia, Palpitation.
- ⑤ Metabolism and nutrition disorders: Creatinekinase (CK) increase, hyperglycemia, hypertriglyceridemia, hyperuricemia
- ⑥ Musculoskeletal disorders: myalgia
- ⑦ Platelets and bleeding: Epistaxis
- ⑧ Psychiatric and nervous disorders: Anxiety, Depression, Somnolence.
- ⑨ Respiratory disorders: Dyspnoea
- ⑩ Skin and subcutaneous tissue disorders: Rash, Sweating increase
- ⑪ Urinary disorders: hematuria

2) Heart failure

The pattern of adverse reactions in patients with heart failure was consistent with the pharmacological action and the patient's health status. In the CHARM clinical trial comparing the 32 mg(n=3,803) of this drug and placebo(n=3,796) group, 21.0% of this drug treatment group and 16.1% of the placebo group discontinued treatment due to the adverse reaction. The common ($\geq 1/100$, $< 1/10$) adverse reactions is hypotension, hyperkalemia, renal impairment, and increased creatinine, urea, and potassium. It is recommended to monitor serum potassium and creatinine levels regularly during and after dose escalation.

- 3) In general, this drug did not have a clinically significant effect on routine laboratory parameters. The level of hemoglobin and hematocrit decreased slightly as with the other Renin-Angiotensin-Aldosterone inhibitors. The increased creatinine, urea, potassium and decreased sodium was observed. The adverse reaction of ALT increase was reported with higher frequency than placebo group (1.3% vs 0.5%). Routine monitoring of laboratory parameters is not required in patients treated with this drug, but long-term monitoring of serum potassium and creatinine levels is recommended in patients with renal impairment.
- 4) The adverse reactions occurred very rarely ($< 1/10,000$) or not known of Post Marketing Surveillance are follows
 - (1) Blood and lymphatic disorders: leukopenia and neutropenia may occur. And agranulocytosis may occur thus after sufficient observation, when abnormality is recognized, stop administration of this drug and proper treatment is taken.
 - (2) Metabolism and nutrition disorders: Hyperkalemia, hyponatremia, hypoglycemia (common in patients being treated for diabetes) may occur thus after sufficient observation, when Asthenia, hunger, cold sweat, hand tremors, poor concentration, convulsions, impaired consciousness appear, stop administration of this drug and proper treatment is taken.
 - (3) Nervous system disorders: Dizziness, headache may occur
 - (4) Gastrointestinal disorders: Nausea, diarrhea may occur
 - (5) Hepatobiliary disorders: Hepatic enzymes increased, hepatitis may occur. And hepatic function impairment such as AST, ALT and γ -GTP increased and jaundice may occur thus after sufficient observation, when abnormality is recognized, stop administration of this drug and proper treatment is taken

- (6) Skin and subcutaneous tissue disorders: Rash, urticaria and pruritus may occur. And angioedema (Swelling in face, mouth, lips, tongue and laryngopharynx) may occur thus after sufficient observation, when abnormality is recognized, stop administration of this drug and proper treatment is taken
 - (7) Rhabdomyolysis characterized by myalgia, asthenia, Creatinekinase (CK) increase, myoglobin increase in blood and urine may occur thus after sufficient observation, when it occurs, stop administration of this drug and proper treatment is taken
 - (8) Musculoskeletal disorders: Myalgia, arthralgia and back pain may occur
 - (9) Renal and urinary disorders: renal impairment, acute renal failure may occur, after sufficient observation, when abnormality is recognized, stop administration of this drug and proper treatment is taken
 - (10) Interstitial pneumonia accompanied by fever, chronic cough dyspnea, and chest X-ray abnormality may occur, thus when it occurs, stop administration of this drug and proper treatment is taken
 - (11) Shock, fainting, loss of consciousness (if chronic renal failure, fainting, loss of consciousness is 0.1~5%): Shock, fainting caused by blood pressure decrease and loss of consciousness may occur, thus observe thoroughly and if cold sensation, vomiting, and loss of consciousness occurs, take proper treatment. Especially, hemodialysis, strict sodium restriction, diuretic administration discontinues etc. and patients with renal failure start admonition with low dose of this drug and in case of increasing the dose, it should be done gradually while fully observing the patient's condition.
- 5) According to the results of Post marketing surveillance for domestic drug re-evaluation in 15,585 subjects, the incidence rate of adverse reactions was reported to be 1.06%(166/15,585 subjects) regardless of causal relationship, of those, 0.82%(128/15,585) were found to have a causal relationship with this drug. Regardless of causal relationship, dizziness the most at 0.30%(47), followed by headache 0.22%(35) and cough 0.17%(26). Among them, the new adverse reactions that did not appear in pre-marketing clinical trials are follows.
- (1) Central and peripheral nervous disorders: Paralysis (2), Nervousness(1)
 - (2) General disorders: Syncope(1), Flushing(1), Weight increase(1), Face edema (1)
 - (3) Gastrointestinal disorders: Thirst (2)
 - (4) Psychiatric and nervous disorders: Sleep disorders(2), Libido decreased (1), Forgetfulness (1), anorexia(2), Appetite increase(1)
 - (5) Sensory disorders: Tinnitus(1), Taste perversion (1)
 - (6) Reproductive disorders: Impotence(1)

13. Symptoms and Treatment of Overdose

There are no study reports about over dosage to human.

13.1. Candesartan cilexetil

Based on pharmacological considerations, the main manifestation of an overdose is likely to be symptomatic hypotension and dizziness. In individual case reports of overdose (of up to 672 mg candesartan cilexetil) patient recovery was uneventful. If hypotension occurs due to this drug overdose, following symptomatic treatments should be instituted while monitoring vital signs. The patients should be placed supine with the legs elevated and plasma volume should be corrected. If this is not sufficient, administer sympathomimetic medical products. This drug cannot be removed by hemodialysis.

13.2. Amlodipine

Severe overdose with amlodipine may result in excessive peripheral vasodilation and, possibly, reflex tachycardia so far based on data. And marked and prolonged systemic hypotension up to and including shock with fatal outcome have been reported. Absorption of amlodipine was significantly decreased when activated charcoal was administered to healthy volunteers immediately after administration of 10 mg of amlodipine or after 2 hours. Gastric lavage may be useful in some cases. The severe hypotension due to overdose require cardiovascular supporting including cardiac and respiratory monitoring frequently and maintain the position of the extremities higher than the body to ensure sufficient amount of circulating body fluids such as blood and urine excretion, etc. A vasoconstrictor may be helpful in restoring vascular tone and blood pressure, except in special cases where the vasoconstrictor is contraindicated. Intravenous calcium gluconate may be beneficial in reversing the effects of calcium channel blockade. Because the protein binding rate is high, the hemodialysis is not useful.

14. Effects on Ability to Drive and Use Machine

It should be taken in to account that occasionally dizziness or weariness may occur during treatment of hypertension because of blood pressure decrease. Thus be caution when drive or operate machine.

15. Preclinical Safety Data

- Amlodipine

Carcinogenesis, Mutagenesis, Impairment of Fertility

Rats and mice treated with amlodipine in the diet for 2 years, at concentrations calculated to provide daily dosage levels of 0.5, 1.25, and 2.5 mg/kg/day showed no evidence of carcinogenicity. The highest dose (for mice, similar to, and for rats twice* the maximum recommended clinical dose of 10 mg, on a mg/m² basis) was close to the maximum tolerated dose for mice but not for rats.

Mutagenicity studies revealed no drug-related effects at either the gene or chromosome levels.

There was no effect on the fertility of rats treated with amlodipine (males for 64 days and females for 14 days prior to mating) at doses up to 10 mg/kg/day (8 times* the maximum recommended human dose of 10 mg, on a mg/m² basis).

*Based on patient weight of 50 kg.

- Candesartan

In a variety of preclinical safety studies conducted in several species, expected exaggerated pharmacological effects, due to modification of the renin-angiotensin-aldosterone system homeostasis, have been observed. The incidence and severity of the effects induced were dose and time related and have been shown to be reversible in adult animals. Animal studies with candesartan cilexetil have demonstrated late foetal and neonatal injury in the kidney. The mechanism is believed to be pharmacologically mediated through effects on the renin-angiotensin-aldosterone system.

There was no evidence of mutagenicity, clastogenicity or carcinogenicity.

The renin-angiotensin-aldosterone system plays a critical role in kidney development. Renin-angiotensin-aldosterone system blockade has been shown to lead to abnormal kidney development in very young mice. Administering drugs that act directly on the renin-angiotensin-aldosterone system can alter normal renal development.

16. Storage Conditions: Store below 25°C

17. Dosage forms and packaging available

Film-coated tablet (candesartan cilexetil/amlodipine): 8/5 mg, 16/5mg

30 Tablets (10 Tablets/PTP x 3)

18. Name and address of manufacturer/Product registration holder

18.1 Manufacturer

HK inno.N Corporation

239, Osongsaengmyeong 2-ro, Osong-eup, Heungdeok-gu, Cheongju-si, Chungcheongbuk-do,
Republic of Korea

18.2 Product Registration Holder

Eucogen Sdn. Bhd.

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19. Date of revision of PI

August 01, 2024